



A Bayesian framework for uncertainty quantification of rainfall Intensity–Duration–Frequency (IDF) curves

Bijoychandra S. Takhellambam^{a,*}, Puneet Srivastava^a, Jasmeet Lamba^b, Hemendra Kumar^a, Roberto Molinari^c

^a University of Maryland, College of Agriculture and Natural Resources, 1201 Symons Hall, College Park, MD, 20742, USA

^b Auburn University, Department of Biosystem Engineering, 350 Mell St, Auburn, AL 36849, USA

^c Auburn University, Department of Mathematics and Statistics, 201 Extension Hall, AL 36849, USA

ARTICLE INFO

Keywords:

Bayesian
Markov-Chain Monte Carlo
Rainfall Extreme
IDF curves
Uncertainty quantification

ABSTRACT

Extreme climate events are projected to occur more frequently in the southeastern United States. However, uncertainties associated with climate studies, particularly those concerning extreme rainfall events, are often neglected due to limitations in climate models, insufficient data, and incomplete understanding of underlying physical processes. This study aims to update and quantify the uncertainties associated with future projected rainfall Intensity–Duration–Frequency (IDF) curves using Bayesian framework under the RCP8.5 scenario. Results indicate that IDF curves developed for the historical baseline (1974–2013) showed lower rainfall intensities compared to NOAA Atlas 14, with differences ranging from –20% to –2%. Comparisons between the Ensemble RCM projections (2030–2059) and the historical baseline revealed that shorter durations showed smaller changes, whereas longer durations exhibited substantial increases in rainfall intensity, indicating significant projected impacts of climate change on extreme rainfall events. The relative changes in rainfall intensities ranged from –43% to 220%. The uncertainty assessment showed rainfall intensity spreads ranging from 1 mm/h to 56 mm/h at the 95% credible interval. Mean rainfall intensities for durations of 0.25, 0.5, 1, 3, 6, 12, and 24 h across the 2-, 5-, 10-, 25-, 50-, and 100-year return periods ranged from 6 mm/h to 177 mm/h.

1. Introduction

Climate change has increased the magnitude and frequency of extreme rainfall events in many parts of the United States, and climate models project that these trends will continue into the 21st century (Lopez-Cantu et al., 2020; Prein et al., 2017). Historically, extreme daily rainfall events of 76.2 mm or more occurred at an estimated average of 0.95 days per year in the early 1900s, but their frequency increased by 23%, 16%, and 20% during the 1990s, 2000s, and 2010s, respectively (Easterling et al., 2017). The lower increasing rate of frequency in 2000s could have been due to the decadal variability, sampling uncertainty rather than weakening of the signal (Deser et al., 2020). Additionally, the IPCC (2018) projected that global temperatures may rise by approximately 1.5 °C between 2030 and 2052 under current greenhouse gas emission rates. As temperatures increase, evaporation rates and atmospheric moisture content also rise (Wang et al., 2016), which is expected to lead to a roughly 7% increase in rainfall intensity for each 1 °C of warming (Easterling et al., 2017). In addition, dynamic effects have

resulted in a decrease in Earth's surface area receiving daily precipitation. The redistribution of precipitation due to changing cloud structures, meteorological phenomena, and atmospheric circulation may lead to more concentrated precipitation shifting towards pole and drought conditions in lower latitudes. This is primarily due to the expansion of low relative humidity zones on the lower to mid troposphere. (Dobler et al., 2024).

The major consequences of extreme events negatively affect the socioeconomic stability of a region. Flood events can be further amplified by urbanization, specifically the increase in paved and impervious surfaces which reduces runoff time and contributes to flash flood events (Zhao et al., 2023, 2026). Developing accurate rainfall Intensity–Duration–Frequency (IDF) curves is therefore essential for designing hydrological and hydraulic structures such as dams, sediment control systems, and stormwater drainage networks, helping to reduce the impacts of extreme rainfall (Almazroui et al., 2019; Ng et al., 2019; Ouali and Cannon, 2018; Tung and Wong, 2014). Existing infrastructure in the United States is largely designed based on historical extreme rainfall

* Corresponding author.

E-mail address: bijoytbl@umd.edu (B.S. Takhellambam).

<https://doi.org/10.1016/j.atmosres.2026.109165>

Received 13 April 2026; Received in revised form 9 June 2026; Accepted 19 June 2026

Available online 20 June 2026

0169-8095/© 2026 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

events, whereas future infrastructure will need to incorporate the projected changes in rainfall extremes (Lopez-Cantu et al., 2020). Research has shown that non-stationarity in rainfall patterns is increasing, and stationary IDF curves tend to underestimate rainfall extremes (Mirhosseini et al., 2014; Sarhadi and Soulis, 2017; Vu and Mishra, 2019; Zhu, 2013; Zhu et al., 2012).

The development of future projected rainfall IDF curves is particularly important for the Southeast United States, a region expected to experience significant changes in rainfall characteristics (McGehee and Srivastava, 2018). Previous studies (Cheng and AghaKouchak, 2014; Ganguli and Coulibaly, 2019; Ghasemi Tousi et al., 2021; Mirhosseini et al., 2013, 2014; Sarhadi and Soulis, 2017; Zhu, 2013; Zhu et al., 2012) have typically used aggregated rainfall data (hourly or daily) to develop non-stationary IDF curves. However, coarse temporal aggregation can smooth rainfall peaks and lead to underestimation of extreme events when compared to breakpoint or sub-hourly rainfall (McGehee and Srivastava, 2018). To address this, studies by Mirhosseini et al. (2013, 2014, 2015) and Takhellambam et al. (2024) developed sub-hourly rainfall-based IDF curves for the Southeast United States while accounting for non-stationarity. For instance, Mirhosseini et al. (2013) employed six global and regional climate models with a 3-h resolution over Alabama and stochastically disaggregated the data to 15-min intervals, developing IDF curves using the generalized extreme value (GEV) distribution. Their findings highlighted the importance of incorporating uncertainty in datasets derived from climate models. Subsequently, Mirhosseini et al. (2014) improved the disaggregation process using a feed-forward backpropagation artificial neural network, producing sub-hourly rainfall data for developing GEV-based IDF curves. Takhellambam et al. (2024) similarly developed projected IDF curves for the Southeast United States using sub-hourly rainfall generated through a modified stochastic method.

In addition to obtaining point estimates for IDF curves, uncertainty quantification (especially for extreme rainfall) is essential for stakeholders such as policymakers, researchers, and farmers engaged in climate adaptation and mitigation (Katz et al., 2013; Guttorp, 2012). It allows for a better understanding of climate model variability and supports the development of more resilient management practices (Ng et al., 2019; Zhao et al., 2022). In the above studies, however, uncertainty associated with IDF curves was quantified using frequentist inference led to higher uncertainty for higher return period. However, Bayesian approach provides more robust as it fits with a true probabilistic model (Chaudhuri and Sharma, 2020).

Additionally, epistemic uncertainties arise from climate models (Al Kajbaf et al., 2023; Emmanouil et al., 2022) due to limited understanding of physical processes, insufficient information, or analytical constraints (Mirhosseini et al., 2015; Solaiman, 2011). Indeed, climate models also depend on assumptions related to the carbon cycle and greenhouse gas concentrations, both of which introduce additional uncertainties (Solaiman, 2011). These uncertainties are further influenced by choices in statistical modeling, including distribution selection and parameter estimation methods (Al Kajbaf et al., 2023). A core challenge is the lack of a universally valid probability model for IDF estimation. Bootstrapping is often applied to mitigate uncertainty, but larger quantiles tend to introduce greater uncertainty (Hailegeorgis et al., 2013). Several studies (Mélèse et al., 2018; Chaudhuri and Sharma, 2020; Coles et al., 2003) have found Bayesian frameworks to be more robust for uncertainty quantification in rainfall IDF curves. Considering these limitations, this study aims to quantify the uncertainties associated with future projected rainfall IDF curves for the eleven largest cities in the Southeastern United States under the RCP8.5 scenario, utilizing sub-hourly rainfall datasets for the period 2030–2059. The parameter uncertainties of the best fit model are assessed within a Bayesian framework based on posterior distributions. Subsequently, the IDF curves are constructed using these posterior distributions, thereby providing a probabilistic characterization of the uncertainty range in the estimated IDF relationships.

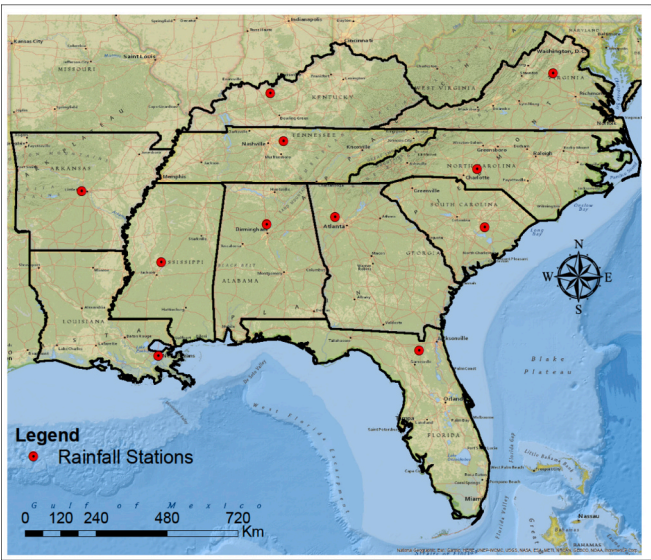


Fig. 1. Map showing the locations of 11 rainfall stations.

Table 1
Descriptions of 11 rainfall station locations.

No.	Name of Sites	State	Longitude (° W)	Latitude (° N)
1	Warrior	Arkansas	92.33	34.74
2	Ferndale 6 E	Louisiana	90.07	29.95
3	Raiford State Prison	Mississippi	90.18	32.30
4	Dallas 7 Ne	Alabama	86.81	33.52
5	Fordsville	Tennessee	86.78	36.17
6	New Orleans Audubon	Kentucky	85.77	38.25
7	Canton 4n	Georgia	84.39	33.75
8	Mount Pleasant	Florida	81.66	30.34
9	Manning	Virginia	75.98	36.85
10	Lebanon	North Carolina	80.84	35.22
11	Piedmont Research Sta	South Carolina	79.93	32.77

2. Data and study area

We selected 11 rainfall stations located nearest to each state's largest city in the Southeast United States—Alabama, Arkansas, Florida, Georgia, Kentucky, Louisiana, Mississippi, North Carolina, South Carolina, Tennessee, and Virginia (Fig. 1 and Table 1). Rainfall in this region ranges from 1000 to 1250 mm per year, which is significantly higher than the national average of 856 mm (Ingram et al., 2013; Takhellambam et al., 2022). The National Oceanic and Atmospheric Administration National Climatic Data Center (NOAA NCEI (National Oceanic and Atmospheric Administration, National Centers for Environmental Information), 2014) provides 15-min measured rainfall data from 1974 to 2013. Quality-controlled stations were selected using the “20.11” screening method, which requires at least 20 years of rainfall records and at least 11 months of complete measurements per year (McGehee and Srivastava, 2018).

Historical (1970–2005) and future (2030–2059) projected rainfall datasets were obtained from five climate models within the North American Coordinated Regional Downscaling Experiment (NA-CORDEX), each with a temporal resolution of 1 h and a spatial resolution of approximately 50 km. The selected models include CANESM (Canadian Earth System Model–Canadian Regional Climate Model version 4), HADGEM (Hadley Centre Global Environment Model version 2 Earth System Model–Weather Research and Forecasting), GFDL (Geophysical Fluid Dynamics Laboratory Earth System Model), MPIREG (Regional Climate Model version 4 driven by the Max Planck Institute for

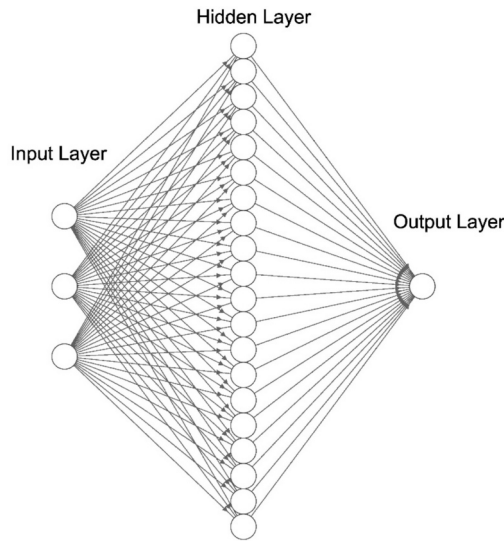


Fig. 2. A multilayer feed forward back propagation perceptron with a hidden layer.

Meteorology Earth System Model LR), and MPIWRF (Weather Research and Forecasting model driven by the Max Planck Institute for Meteorology Earth System Model LR) (Giorgi and Anyah, 2012; Scinocca et al., 2016; Skamarock et al., 2005). All models were forced under the RCP8.5 scenario. This datasets represent spatial average for a domain and is not directly equivalent to rainfall observations. Therefore, rainfall datasets were further extracted for a given location and were bias-corrected following the approach of Takhellambam et al. (2022). Although, bias correction reduce systematic biases and become standard procedure, there are few studies that questions the rationale of using it as it cannot fully compensate for limitations arising from violating conservation principles, altering spatiotemporal field consistency, and unresolved physical processes (Ehret et al., 2012; Muerth et al., 2013). Therefore, there are some uncertainty associated with the model's minimum skillful scale in the rainfall datasets.

3. Methodology

Rainfall datasets from both hourly observed station records and bias-corrected climate model simulations future rainfall (30 years i.e., 2030–2059) were acquired and pre-processed to quantify uncertainties associated with the developed rainfall IDF curves. The workflow included disaggregating bias-corrected hourly rainfall into sub-hourly monthly maximum values and subsequently developing IDF curves. The detailed procedures are described in the following subsections.

3.1. Rainfall disaggregation using neural networks

Multilayer perception (MLP) models were developed for the temporal disaggregation of monthly maximum hourly rainfall into sub-hourly monthly maximum rainfall at 15-min and 30-min intervals. The model architecture consists of an input layer, a hidden layer with 20 neurons, and an output layer (Fig. 2). To estimate extreme rainfall at finer temporal resolutions from 1-h values, the input layer incorporates the concurrent, preceding, and succeeding hourly rainfall values rather than the full rainfall time series. The rectified linear unit (ReLU) activation function was used, with mean squared error as the loss function and root mean square propagation (RMSprop) as the optimizer. The model was trained using the Adam optimizer on 70% of the dataset, while the remaining 30% was used for testing. Readers are referred to Takhellambam et al. (2024) for details.

3.2. Bayesian inference framework for development of rainfall intensity duration frequency curves

3.2.1. Gumbel distribution and IDF curves

The observed 15-min rainfall datasets were aggregated to 0.5-h (30-min), 1-h, 3-h, 6-h, 12-h, and 24-h durations. Similarly, the Ensemble RCM rainfall data at 1-h intervals were aggregated to longer durations. Using these aggregated datasets, along with the monthly maximum rainfall generated by the NN model, annual maximum rainfall series were extracted for durations of 0.25 h (15-min), 0.5 h (30-min), 1 h, 3 h, 6 h, 12 h, and 24 h.

Mirhosseini et al. (2014) and Takhellambam et al. (2024) recommended the Type-I Generalized Extreme Value (GEV) distribution, also known as the Gumbel distribution (shape parameter = 0) for developing IDF curves in the Southeast United States. The probability distribution function (PDF) of the Gumbel distribution is defined as:

$$G(x) = \exp \left[-\exp \left\{ -\left(\frac{x - \mu}{\sigma} \right) \right\} \right] \quad (1)$$

where x is the annual maximum rainfall intensity, while μ and σ are location and scale, respectively.

Several methods exist for estimating the parameters of the Gumbel distribution, including maximum likelihood estimation (MLE), L-moments, and the method of moments (Durrans and Brown, 2001; Hosking et al., 1985). Previous studies have found MLE to be superior for the Gumbel distribution for the region because it assumes the model is well specified and provides optimal statistical properties compared to other techniques such as the method of moments (Mahdi and Cenac, 2005). However, MLE can exhibit large uncertainties when estimating extreme events, especially for short-duration rainfall. Although confidence intervals (e.g., 95%) may be constructed for the Gumbel distributions, these intervals are often wide, especially for large quantiles, leading to additional uncertainty (Chaudhuri and Sharma, 2020).

With the point estimates of parameters of the Gumbel distribution using MLE, the IDF curves for rainfall depth, denoted as X_T , for a given return periods (T) say 2, 5, 25, 50, and 100-years were developed using the Gumbel distribution as given below:

$$X_T = \bar{x} - \sigma \ln \left(\ln \left(\frac{T}{T-1} \right) \right) \quad (2)$$

where, $\bar{x}$ and σ are the mean and standard deviation of rainfall, respectively.

3.2.2. Uncertainty in IDF curves

To address the limitations with point estimates of parameters, we used Bayesian inference to obtain the full posterior probability distribution of the location and scale parameters of the Gumbel distribution. This approach enables explicit quantification of uncertainty associated with the estimated IDF curves and allows for the integration of prior knowledge.

By Bayes Theorem, the posterior distribution $P(\theta|X)$ of the parameters, which is represented by $\{(\mu, \sigma) : \theta\}$ is defined as:

$$P(\theta|X) \propto L(X|\theta) P(\theta) \quad (3)$$

Where, $L(X|\theta)$ and $P(\theta)$ are the likelihood function and prior distribution for datasets X , respectively.

This posterior distribution of parameters are estimated using Markov Chain Monte Carlo (MCMC) method, particularly Metropolis-Hastings algorithm (Cotton and Ferro, 2008). This algorithm can be described as:

- Start with an initial value of θ_0 which is obtained using MLE of both location and scale parameters. In our study, we assumed a non-informative normal prior for both location and scale parameter with large variance say 10,000 (Cotton and Ferro, 2008; Vidal, 2014).

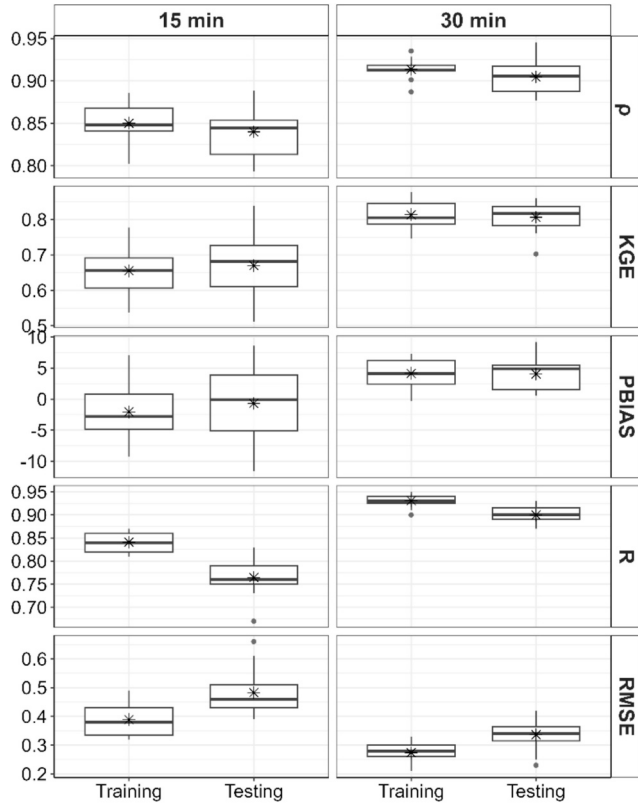


Fig. 3. Boxplot showing the overall performance of 11 rainfall stations for disaggregation of monthly maximum 1-h to 15 min and 30 min. Asterisks (*) denote the average value.

- b) With random walk Metropolis Hastings method, the proposal values of parameters are generated with

$$\theta^* \sim N(\theta_i, \sigma^2) \quad (4)$$

Where, $i = 0, 1, 2, 3, \dots$ and specified standard deviation (σ).

- c) The accept-reject for proposed parameter value:

$$\text{Acceptance probability, } r = \min\left\{1, \frac{p(\theta^*)}{p(\theta_i)}\right\} \quad (5)$$

where, $p(\theta^*)$ and $p(\theta_i)$ are arbitrary probability distribution.

$$\text{New parameters, } \theta_{i+1} = \begin{cases} \theta^* & \text{if } u \leq r \\ \theta_i & \text{otherwise} \end{cases} \quad (6)$$

where u is uniformly distributed random number between 0 and 1.

After a burn-in period, which discarded large sample of parameters, the remaining θ become dependent sample form the posterior distribution. Using this sequence of both location and scale parameter, the IDF curves were estimated under Gumbel Distribution. Furthermore, the credible intervals which are similar to confidence interval are obtained for significance level at 95%.

3.3. Statistical analysis

We use the following statistical metrics to evaluate the performance of rainfall disaggregation.

$$\text{Pearson correlation, } r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}} \quad (7)$$

$$\text{Root mean square error, } RMSE = \sqrt{\frac{\sum (x_i - y_i)^2}{n}} \quad (8)$$

$$\text{Kling - Gupta Efficiency, } KGE = 1 - \sqrt{(r - 1)^2 + (\alpha - 1)^2 + (\beta - 1)^2} \quad (9)$$

$$\text{Where, } \alpha = \frac{\sigma_y}{\sigma_x} \text{ are } \beta = \frac{\mu_y}{\mu_x}$$

$$\text{Percent Bias, } PBIAS = \frac{\sum (x_i - y_i)}{\sum x_i} \times 100\% \quad (10)$$

$$\text{Spearman Rank Correlation, } \rho = 1 - \frac{6 \sum d^2}{n(n^2 - 1)} \quad (11)$$

where, x and y are the observed and simulated rainfall and n number of data points. $\bar{x}$ and $\bar{y}$ are mean rainfall values of observed and simulated rainfall. d is the difference between the rank of simulated and observed rainfall.

4. Results and discussion

4.1. Performance of rainfall disaggregation

The observed 15-min rainfall data were aggregated to 1-h rainfall. From both datasets, the monthly maximum rainfall values at 15-min and 1-h intervals were extracted. The observed monthly maximum 1-h rainfall was then disaggregated using the NN model to obtain simulated 15-min rainfall datasets. The performance of the NN model was evaluated by comparing the observed and simulated monthly maximum rainfall. The same procedure was used to generate simulated 30-min monthly maximum rainfall.

Fig. 3 presents the performance of the rainfall disaggregation across all 11 rainfall stations. During training period of 15-min, the average KGE, r , and ρ values are greater than 0.65. The ranges of KGE, r , and ρ varies from 0.53 to 0.7, 0.81 to 0.87 and 0.8 to 0.88, respectively. Whereas the average RMSE and PBIAS are 0.39 and -2% , respectively. The range of RMSE and PBIAS are 0.32 to 0.49 and -9.2% to 7.1% , respectively.

During testing period, the average KGE, r , and ρ values are 0.67, 0.76, and 0.84, respectively. In addition, testing period has an averaged RMSE and PBIAS of 0.46 and -0.6% , respectively.

However, the performance of 30-min is better than that of 15-min disaggregation. Training period shows an average KGE, r , and ρ values are greater than 0.81. The ranges of KGE, r , and ρ varies from 0.75 to 0.87, 0.9 to 0.95 and 0.88 to 0.93, respectively. Whereas the average RMSE and PBIAS are 0.27 and 4% , respectively. The range of RMSE and PBIAS are 0.21 to 0.33 and -0.3% to 7.3% , respectively. Similarly, the averaged value during testing period of KGE, r , and ρ are greater than 0.8. In addition, the average RMSE and PBIAS are 0.33 and 4% , respectively.

One of the main reasons for the improved performance of the 30-min NN model is the smoothing effect of rainfall peaks associated with longer temporal aggregation, which reduces variability and enhances model stability. Overall, the performance of the NN models for rainfall disaggregation is considered satisfactory and aligns with the findings of Mirhosseini et al. (2014), who recommended a Nash–Sutcliffe Efficiency greater than 0.6.

4.2. Rainfall IDF Curves And Uncertainty

4.2.1. Historical baseline with comparison to NOAA Atlas 14

Rainfall IDF curves were developed for the historical baseline using NOAA rainfall datasets (1974–2013) and were compared with the NOAA Atlas 14 rainfall frequency estimates (Perica et al., 2013). Additionally, rainfall intensities derived from the ensemble of five climate models for

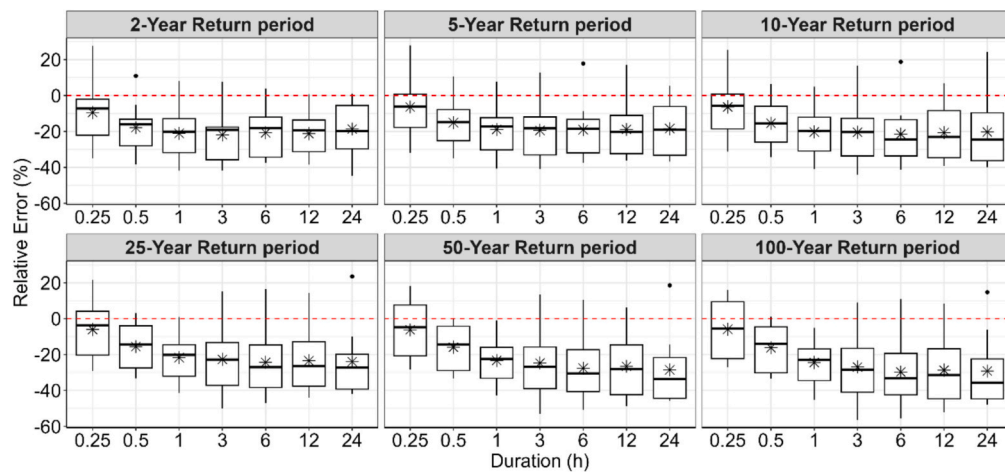


Fig. 4. Comparison of rainfall intensity between historical baseline and NOAA Atlas 14 using relative error. Red dotted line represent the perfect line. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

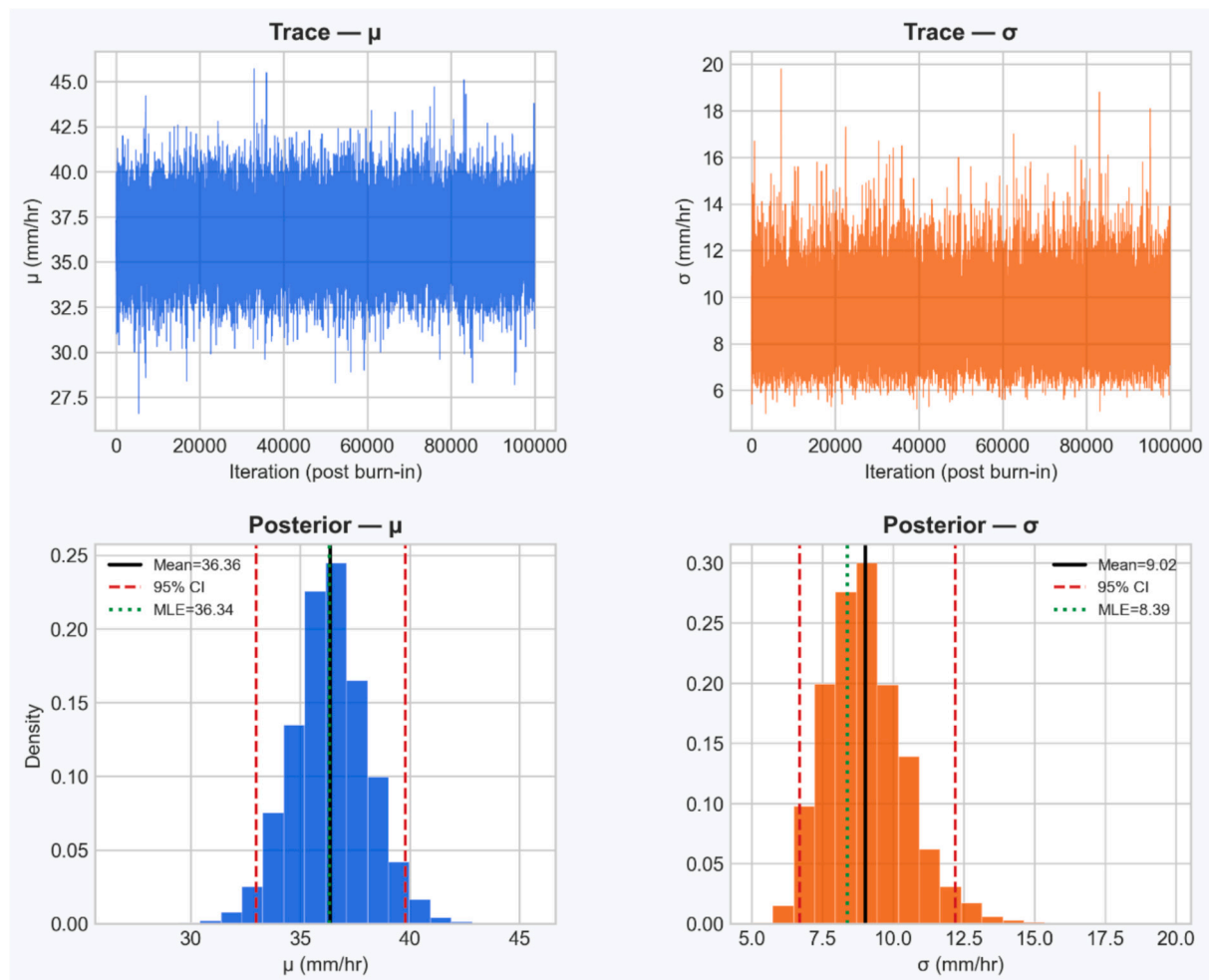


Fig. 5. Trace plot and posterior distribution for both location and scale parameter for rainfall station located at Fordsville, TN of 1 h.

the future period (2030–2059), hereafter referred to as the Ensemble RCM, are discussed.

Fig. 4 shows the overall relative error in rainfall intensity for durations of 15-min (0.25 h), 30-min (0.5 h), 1, 3, 6, 12, and 24 h across return periods of 2, 5, 10, 25, 50, and 100 years. For a given return period, the average rainfall intensities for the 15-min duration under the

historical baseline were close to the corresponding NOAA Atlas 14 values. However, rainfall intensities for all other durations and return periods exhibited underestimation relative to NOAA Atlas 14, with average differences ranging from -20% to -2% . This discrepancy arises because [Perica et al. \(2013\)](#) used the L-moments method to estimate distribution parameters, whereas our study employed MLE.

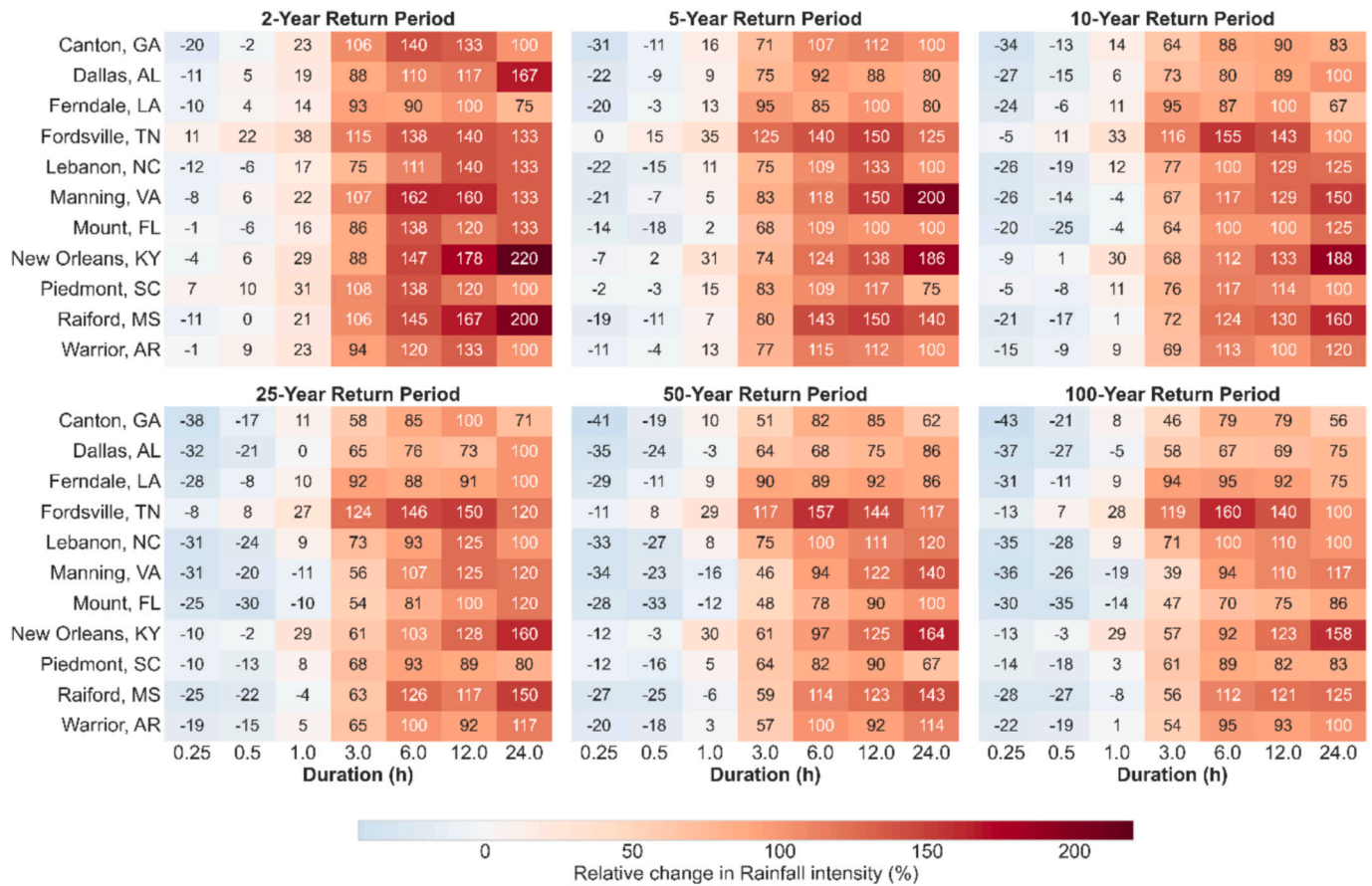


Fig. 6. Heatmap comparing the rainfall intensities for historical baseline and Ensemble RCM for all stations with duration 0.25 (15-min), 0.5 (30-min), 1, 3, 6, 12, and 24 h at return periods of 2, 5, 10, 25, 50, and 100-Year.

Furthermore, both studies used different rainfall datasets, which also contributes to the observed differences.

4.2.2. Uncertainty in rainfall intensity of Ensemble RCM

We expect significant differences in the IDF curves between the Ensemble RCM and the historical baseline for all durations and return periods. This is primarily due to the differences in the underlying rainfall datasets: the Ensemble RCM represents future climate simulations under the worst-case scenario (RCP8.5, with a radiative forcing of 8.5 W/m^2), while the historical baseline is based on observed rainfall. Comparing these two scenarios provides insight into the impacts of climate change on extreme rainfall behavior and helps quantify the magnitude of future extreme events under a high-emission pathway.

We simulated 150,000 iterations using Markov Chain Monte Carlo with a burn-in period of 50,000. Fig. 5 shows Trace plots (top row) and posterior distribution (bottom row) of both location and scale parameters for rainfall station located Fordville, TN for rainfall interval 1 h. The mean value of location parameter is closely related to the MLE value whereas there is a 7.5% increase in scale parameter as compared to the MLE value. The mean value of location parameter shows 36.36 mm/h with the corresponding 95% credible intervals span from approximately 33 mm/h to 40 mm/h. In addition, the scale parameter has mean value of 9 mm/h with a 95% credible interval of 6.7 mm/h and 12 mm/h, respectively.

We developed the rainfall IDF curves using these parameters for all rainfall stations. The rainfall intensity for all return values is extrapolated from finite 30-year climate-model time slices and should be interpreted together with their posterior uncertainty ranges. Fig. 6 shows overall relative changes in rainfall intensities with Ensemble RCM as compared to historical baseline. Results indicate that short-duration

rainfall events generally exhibit lower intensities. Whereas the longer duration events show noticeably greater rainfall intensities in the Ensemble RCM relative to the current scenario. Our results may reflect regional and methodological factors. In particular, the Southeast United States has strong geographic variability such as coastal moisture effects and tropical systems. In addition, short-duration extremes are more sensitive, which may not be fully represented by the RCM resolution and also be affected by the bias correction and disaggregation steps. The overall relative changes in rainfall intensities between the Ensemble RCM and historical baseline ranges from -43% to 220% . The smallest and greatest relative changes are observed in event 100-year return period with duration of 15-min and 24-h at Canton, GA and New Orleans, KY, respectively.

Furthermore, Fig. 7 shows the uncertainty ranges of annual extreme rainfall values for all durations across the return periods of 2, 5, 10, 25, 50, and 100 years under the Ensemble RCM. The smallest and greatest rainfall intensities are 6 mm/h (100-year return period for the 24-h duration) and 177 mm/h (100-year return period for the 15-min duration) at Piedmont, SC, and New Orleans, KY, respectively. However, the minimum and maximum uncertainty range of rainfall intensities from their mean values are 1 mm/h and 56 mm/h at the 95% credible interval. Furthermore, Fig. 8 shows the IDF curves with their 95% credible interval for a rainfall station located at Fordville, TN.

These findings are consistent with those of previous studies. Moore et al. (2015) reported increases in 24-h extreme rainfall events across much of the Southeast United States in most months of the year. Gori et al. (2022) similarly reported a projected 123% increase in 24-h extreme precipitation for a 100-year storm event at a point location in Houston, Texas. Extreme rainfall patterns vary across the region, with the eastern and western parts experiencing extremes during warmer and

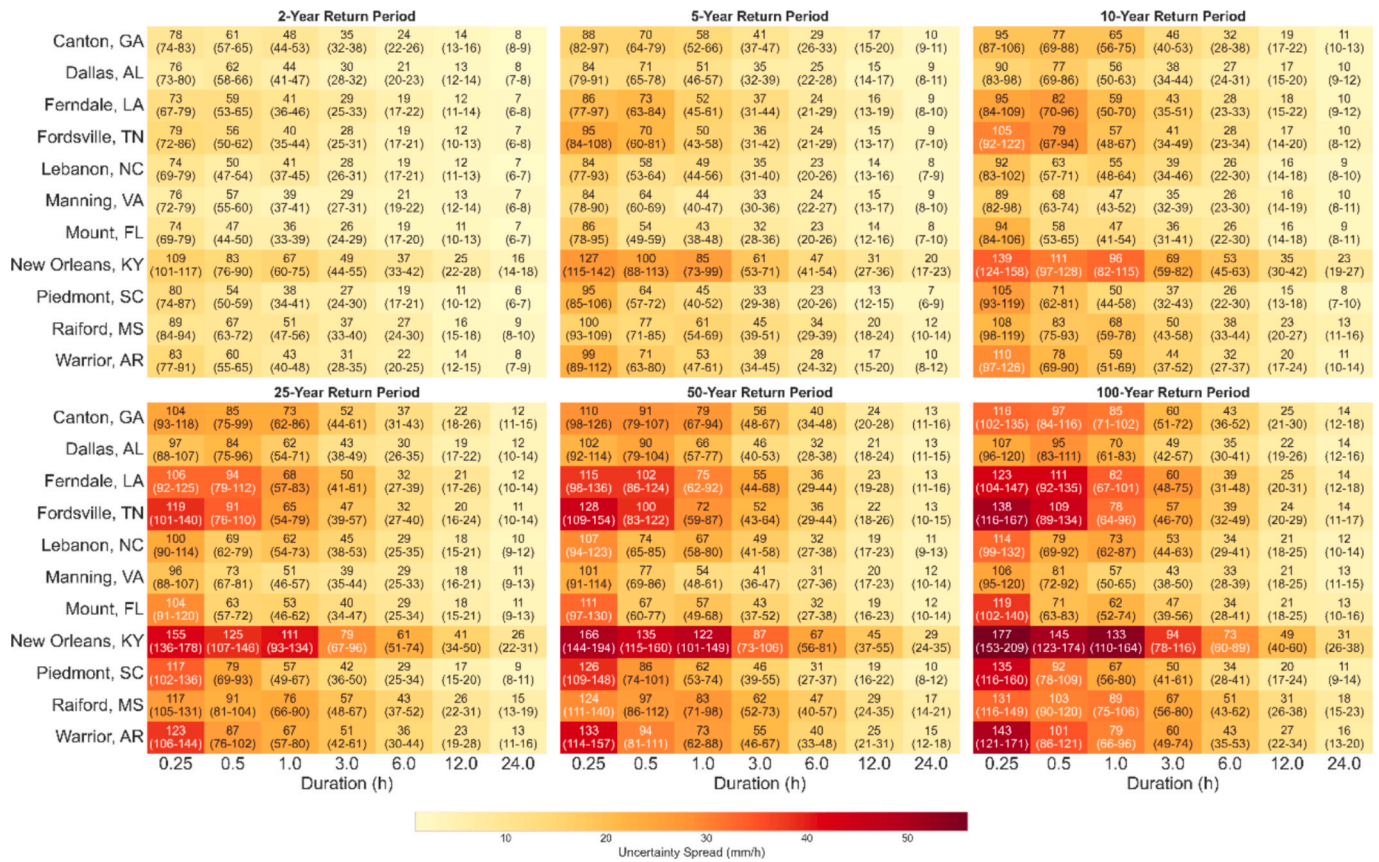


Fig. 7. Uncertainty range of rainfall intensities for durations of 0.25 (15-min), 0.5 (30-min), 1, 3, 6, 12, and 24 h with return period of 2, 5, 10, 25, 50, and 100-year using Ensemble RCM.

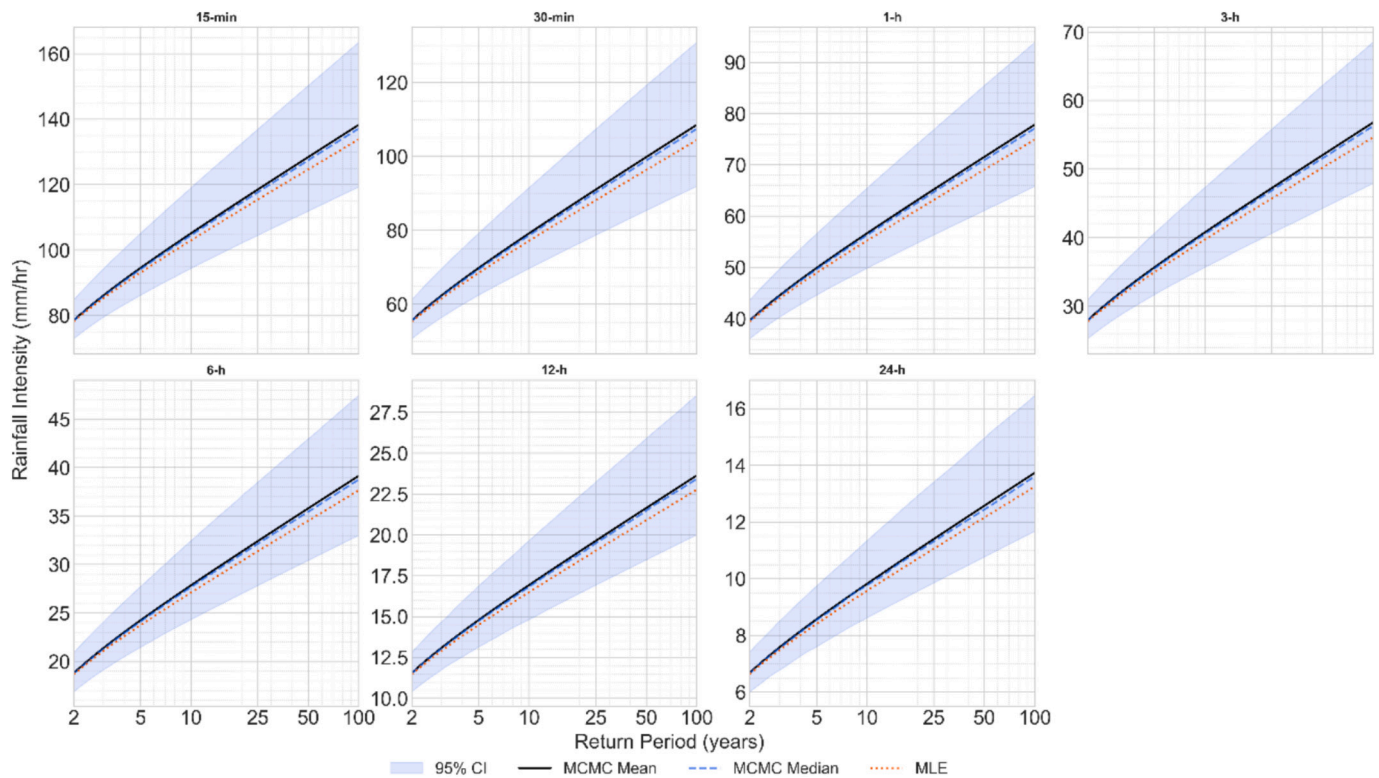


Fig. 8. Rainfall IDF curves comparing between historical baseline and Ensemble RCM for the rainfall station located at Fordville, TN.

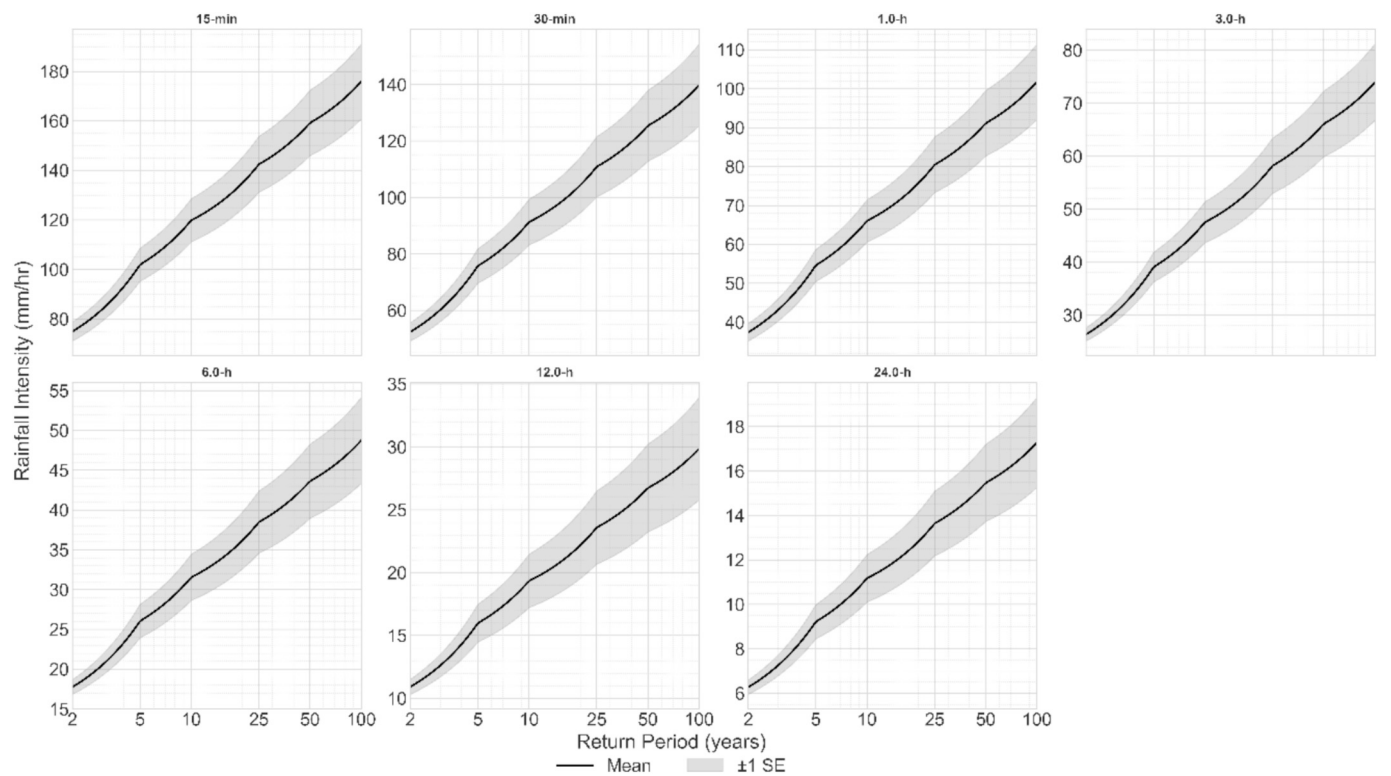


Fig. 9. Inter-model variations in rainfall IDF curves (± 1 Standard error) located at Fordville, TN.



Fig. 10. Heatmap showing relative error due to bias correction on IDF curves.

colder seasons, respectively. Atmospheric conditions influenced by the Atlantic Ocean, Gulf of Mexico, and Appalachian Mountains enhance convective and/or orographic precipitation, contributing to extreme rainfall events across the Southeast. Consistent with expected hydrologic behavior, rainfall intensities generally increase with longer return periods, as higher return periods correspond to rarer and more extreme events associated with larger precipitation volumes.

Furthermore, there are various reasons that could have contributed to the uncertainties in IDF curves. Here, we discussed three main factors

that contributed to the IDF uncertainties i.e., small ensemble size, bias correction, and temporal downscaling.

In this study, we utilized ensemble size of five RCMs, which represents a smaller size compared to recommended practice in climate projection studies, where ensembles of 10–100 members are generally sufficient depending on location, spatial and temporal scale, and quantity of interest (Deser et al., 2020). The use of five RCMs in this study is attributable to the high temporal resolution requirements of the analysis and the limited availability of CORDEX RCMs providing hourly

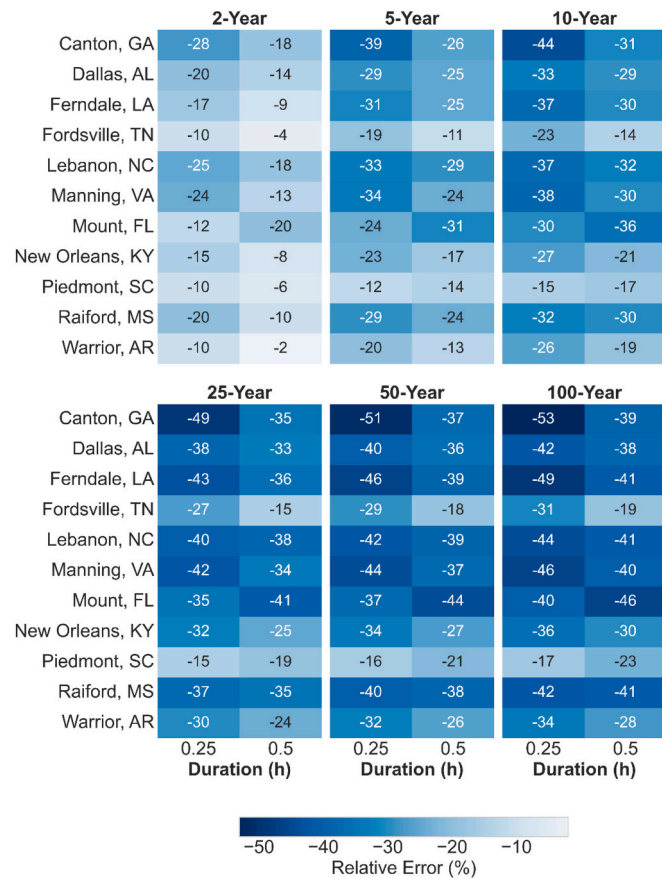


Fig. 11. Heat map showing relative error in IDF curves due to temporal downscaling.

datasets. To quantify the uncertainty associated with the small ensemble size, the standard error (SE) was computed to characterize inter-model spread in IDF curves. Fig. 9 presents inter model variability for rainfall intensity values across all five RCMs for the station located at Fordsville, TN, with ± 1 SE bounds. The overall ranges of SE are 0.2 to 16.4 mm/h and the internal variability increases with increasing return period for a given duration, reflecting greater inter-model divergence at rarer events, a pattern consistent with the known difficulty of constraining extreme precipitation tails with small ensembles (Rampal et al., 2025; Tebaldi et al., 2021). However, this internal variability may be larger than the spread of these five RCMs. Future studies could address by leveraging large multi-model ensembles downscale through empirical-statistical or AI/ML-base emulation methods.

Similarly, Fig. 10 shows the impact of bias correction on the resulting IDF curves. Takhellambam et al. (2022) discussed that bias correction can lead to overestimation of extreme rainfall events, particularly due to the tendency of NA-CORDEX datasets to produce more wet days with low-intensity rainfall. Although the total annual rainfall remains similar, this increase in wet-day frequency significantly affects rainfall characteristics, including the distribution of extremes (Takhellambam et al., 2022). The reason why historical climate models often show lower extreme rainfall values compared to observations is due to area-average quantities in climate models whereas in-situ rain gauge represents point observations. Thus, spatial average lowers the extreme rainfall than the localized precipitation peak values. Furthermore, the minimum skillful scale associated with climate models lacks physical meaning. These structural limitations and statistical mismatch are widely addressed using bias correction.

Additionally, each climate model employs different internal mechanisms, particularly cloud parametrization schemes and model physics,

to simulate rainfall processes. As a result, systematic differences exist between bias-corrected and raw historical datasets for any given model. These inconsistencies suggest that the choice of the bias-correction method can meaningfully influence the annual maximum series used to develop IDF curves. We found the bias correction has resulted in mixed results on the generated IDF curves i.e., both overestimation and underestimation depending on the geographical variability and return period. The overestimation of IDF curves is found with return period less than 10-year with a maximum up to 43%. While there are underestimations of IDF curves due to bias correction in large return period i.e., greater than 10-year return period. The overall associated relative error of IDF curves due to bias correction ranges from -16% to 43%.

Lastly, the temporal downscaled sub-hourly rainfall at 15-min and 30-min has resulted in underestimation of rainfall intensities (Fig. 11). The rainfall intensities for return periods 2-, 5-, 10-, 25-, 50-, and 100-year range from -53% to -2%. Underestimation is further worsened with higher return period. However, longer duration shows a lesser underestimation compared to short duration. This systematic underestimation is primarily because neural network models have poor performance with the downscaling for peak rainfall intensities which is a well-documented limitation.

Moreover, our study is associated with various limitations. First, the grid box climate models are spatially average over the model domain and do not represent the physical real-world conditions. This resulted in the smoothening of the extreme rainfall events resulting in the underestimation of peak events. Secondly, bias correction corrected the climate variables independently while inter-variable dependence is not considered in the procedure. This inter variable dependence is one of the most important in climate impact studies, even misrepresented by climate models.

5. Conclusions

The Southeast United States has significant projected changes in rainfall characteristics, including frequency, intensity, and total amount, driven by climate change as global mean temperatures are projected to increase by 1.5 °C between 2030 and 2052. Climate impact studies and quantifying uncertainty provides valuable information for decision-makers developing adaptation and mitigation strategies.

In this study, we quantify the uncertainties associated with updated rainfall Intensity–Duration–Frequency (IDF) curves for the largest cities in each state of the Southeast United States based on Bayesian inference with the Gumbel distribution. We applied the Metropolis-Hastings algorithm Markov Chain Monte Carlo (MCMC) method with the Hastings algorithm to quantify uncertainties in extreme rainfall. The main findings are:

- Comparison of IDF curves based on historical baseline with NOAA Atlas 14 showed an underestimation ranging from -20% to -2%, which reflects differences in parameter estimation methods and rainfall datasets. In addition, comparisons between Ensemble RCM (2030–2059) and the historical baseline revealed that shorter durations showed lower changes, whereas longer durations exhibited substantial increases in rainfall intensity, confirming that projected climate models indicate significant impacts on extreme rainfall events. The relative changes in rainfall intensities range from -43% to 220%.
- The developed IDF curves based on Ensemble RCM are found with minimum and maximum rainfall intensities of 6 mm/h and 177 mm/h, respectively, for durations of 0.25, 0.5, 1, 3, 6, 12, and 24 h with return periods of 2, 5, 10, 25, 50, and 100 years. Moreover, the uncertainty assessment of rainfall intensities showed ranged from 1 mm/h to 56 mm/h at the 95% credible interval.
- The associated uncertainty was further evaluated from three major sources due to small ensemble size, bias correction, and temporal downscaling. We found there is a internal variability of rainfall

intensities due to difference RCM of 0.2 to 16.4 mm/h. Bias correction of rainfall has resulted in both underestimation and overestimation depending on spatial variability ranging from −16% to 43%. Temporal downscales have also resulted in underestimation of rainfall −53% to −2%.

Considering the associated limitations with this study, future work could work on leveraging large multi-model ensembles downscales through empirical-statistical or AI/ML-base emulation methods on IDF curves. In addition, this framework can be applied to other regions, and additional studies could further improve uncertainty quantification for climate-induced changes in IDF curves.

CRediT authorship contribution statement

Bijoychandra S. Takhellambam: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Puneet Srivastava:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization. **Jasmeet Lamba:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization. **Hemendra Kumar:** Methodology, Conceptualization. **Roberto Molinari:** Writing – review & editing, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was completed in part with resources provided by the Auburn University Hopper Cluster. We are grateful for the support of the USDA-NIFA Hatch Project (ALA014-1-19052), and the Alabama Agricultural Experiment Station for this study.

Data availability

Data will be made available on request.

References

- Al Kajbaf, A., Bensi, M., Brubaker, K.L., 2023. Drivers of uncertainty in precipitation frequency under current and future climate – application to Maryland, USA. *J. Hydrol.* 617, 128775. <https://doi.org/10.1016/j.jhydrol.2022.128775>.
- Almazroui, M., Şen, Z., Mohorji, A.M., Islam, M.N., 2019. Impacts of climate change on water engineering structures in arid regions: case studies in Turkey and Saudi Arabia. *Earth Syst. Environ.* 3, 43–57. <https://doi.org/10.1007/s41748-018-0082-6>.
- Chaudhuri, R.R., Sharma, P., 2020. Addressing uncertainty in extreme rainfall intensity for semi-arid urban regions: case study of Delhi, India. *Nat. Hazards* 104, 2307–2324. <https://doi.org/10.1007/s11069-020-04273-5>.
- Cheng, L., AghaKouchak, A., 2014. Nonstationary precipitation intensity-duration-frequency curves for infrastructure design in a changing climate. *Sci. Rep.* 4, 7093. <https://doi.org/10.1038/srep07093>.
- Coles, S., Pericchi, L.R., Sisson, S., 2003. A fully probabilistic approach to extreme rainfall modeling. *J. Hydrol.* 273, 35–50. [https://doi.org/10.1016/S0022-1694\(02\)00353-0](https://doi.org/10.1016/S0022-1694(02)00353-0).
- Cotton, J., Ferro, C., 2008. *Bayesian Priors and Estimation of Extreme Rainfall in Camborne*. University of Exeter, Cornwall, UK.
- Deser, C., Lehner, F., Rodgers, K.B., Ault, T., Delworth, T.L., DiNezio, P.N., Fiore, A., Frankignoul, C., Fyfe, J.C., Horton, D.E., Kay, J.E., Knutti, R., Lovenduski, N.S., Marotzke, J., McKinnon, K.A., Minobe, S., Randerson, J., Screen, J.A., Simpson, I.R., Ting, M., 2020. Insights from Earth system model initial-condition large ensembles and future prospects. *Nat. Clim. Chang.* 10 (4), 277–286. <https://doi.org/10.1038/s41558-020-0731-2>.
- Dobler, A., Benestad, R.E., Lussana, C., Landgren, O., 2024. CMIP6 models project a shrinking precipitation area. *npj Clim. Atmos. Sci.* 7 (1), 239. <https://doi.org/10.1038/s41612-024-00794-z>.
- Durrans, S., Brown, P.A., 2001. Estimation and internet-based dissemination of extreme rainfall information. *Transp. Res. Rec.* 1743, 41–48. <https://doi.org/10.3141/1743-06>.
- Easterling, D.R., Arnold, J.R., Knutson, T., Kunkel, K.E., LeGrande, A.N., Leung, L.R., Vose, R.S., Waliser, D.E., Wehner, M.F., 2017. Ch. 7: Precipitation Change in the United States. In: *Climate Science Special Report: Fourth National Climate Assessment, Volume I*. U.S. Global Change Research Program. <https://doi.org/10.7930/J0H993CC>.
- Ehret, U., Zehe, E., Wulfmeyer, V., Warrach-Sagi, K., Liebert, J., 2012. HESS Opinions “Should we apply bias correction to global and regional climate model data?”. *Hydrol. Earth Syst. Sci.* 16 (9), 3391–3404. <https://doi.org/10.5194/hess-16-3391-2012>.
- Emmanouil, S., Langousis, A., Nikolopoulos, E.I., Anagnostou, E.N., 2022. The spatiotemporal evolution of rainfall extremes in a changing climate: a CONUS-wide assessment based on multifractal scaling arguments. *Earth's Future* 10, e2021EF002539. <https://doi.org/10.1029/2021EF002539>.
- Ganguli, P., Coulibaly, P., 2019. Assessment of future changes in intensity-duration-frequency curves for Southern Ontario using North American (NA)-CORDEX models with nonstationary methods. *J. Hydrol.: Reg. Stud.* 22, 100587. <https://doi.org/10.1016/j.ejrh.2018.12.007>.
- Ghasemi Tousi, E., O'Brien, W., Doulabian, S., Shadmehri Toosi, A., 2021. Climate changes impact on stormwater infrastructure design in Tucson Arizona. *Sustain. Cities Soc.* 72, 103014. <https://doi.org/10.1016/j.scs.2021.103014>.
- Giorgi, F., Anyah, R.O., 2012. The road towards RegCM4. *Clim. Res.* 52, 3–6. <https://doi.org/10.3354/cr01089>.
- Gori, A., Lin, N., Xi, D., Emanuel, K., 2022. Tropical cyclone climatology change greatly exacerbates US extreme rainfall–surge hazard. *Nat. Clim. Chang.* 12, 171–178. <https://doi.org/10.1038/s41558-021-01272-7>.
- Guttorp, P., 2012. Climate statistics and public policy. *Stat. Polit. Policy* 3. <https://doi.org/10.1515/2151-7509.1055>.
- Hailegeorgis, T.T., Thorolfsson, S.T., Alfredsen, K., 2013. Regional frequency analysis of extreme precipitation with consideration of uncertainties to update IDF curves for the city of Trondheim. *J. Hydrol.* 498, 305–318. <https://doi.org/10.1016/j.jhydrol.2013.06.019>.
- Hosking, J.R.M., Wallis, J.R., Wood, E.F., 1985. Estimation of the generalized extreme-value distribution by the method of probability-weighted moments. *Technometrics* 27, 251–261. <https://doi.org/10.1080/00401706.1985.10488049>.
- Ingram, K., Dow, K., Carter, L., Anderson, J., 2013. *Climate of the Southeast United States: Variability, Change, Impacts, and Vulnerability*. Island Press, Washington DC.
- IPCC, 2018. Global warming of 1.5°C: An IPCC special report on the impacts of global warming of 1.5°C above pre-industrial levels and related global greenhouse gas emission pathways. In: *The Context of Strengthening the Global Response to the Threat of Climate Change, Sustainable Development, and Efforts to Eradicate Poverty*. Cambridge University Press, Cambridge, UK and New York, NY, USA. <https://doi.org/10.1017/9781009157940>.
- Katz, R.W., Craigmiller, P.F., Guttorp, P., Haran, M., Sansó, B., Stein, M.L., 2013. Uncertainty analysis in climate change assessments. *Nat. Clim. Chang.* 3, 769–771. <https://doi.org/10.1038/nclimate1980>.
- Lopez-Cantu, T., Prein, A.F., Samaras, C., 2020. Uncertainties in future U.S. extreme precipitation from downscaled climate projections. *Geophys. Res. Lett.* 47, e2019GL086797. <https://doi.org/10.1029/2019GL086797>.
- Mahdi, S., Cenac, M., 2005. Estimating parameters of Gumbel distribution using the methods of moments, probability weighted moments and maximum likelihood. *Rev. Mat.* 12, 151–156. <https://doi.org/10.15517/rmta.v12i1-2.259>.
- McGehee, R., Srivastava, P., 2018. Benchmarking reliable erosion indices from quarter-hour station data for climate studies in the southeastern United States. *J. Soil Water Conserv.* 73, 363–376. <https://doi.org/10.2489/jswc.73.4.363>.
- Méleze, V., Blanchet, J., Molinié, G., 2018. Uncertainty estimation of intensity-duration-frequency relationships: a regional analysis. *J. Hydrol.* 558, 579–591. <https://doi.org/10.1016/j.jhydrol.2017.07.054>.
- Mirhosseini, G., Srivastava, P., Stefanova, L., 2013. The impact of climate change on rainfall Intensity–Duration–Frequency (IDF) curves in Alabama. *Reg. Environ. Chang.* 13, 25–33. <https://doi.org/10.1007/s10113-012-0375-5>.
- Mirhosseini, G., Srivastava, P., Fang, X., 2014. Developing rainfall intensity-duration-frequency curves for Alabama under future climate scenarios using artificial neural networks. *J. Hydrol. Eng.* 19, 04014022. [https://doi.org/10.1061/\(ASCE\)HE.1943-5584.0000962](https://doi.org/10.1061/(ASCE)HE.1943-5584.0000962).
- Mirhosseini, G., Srivastava, P., Sharifi, A., 2015. Developing probability-based IDF curves using Kernel density estimator. *J. Hydrol. Eng.* 20, 04015002. [https://doi.org/10.1061/\(ASCE\)HE.1943-5584.0001160](https://doi.org/10.1061/(ASCE)HE.1943-5584.0001160).
- Moore, B.J., Mahoney, K.M., Sukovich, E.M., Cifelli, R., Hamill, T.M., 2015. Climatology and environmental characteristics of extreme precipitation events in the Southeastern United States. *Mon. Weather Rev.* 143, 718–741. <https://doi.org/10.1175/MWR-D-14-00065.1>.
- Muerth, M.J., Gauvin St-Denis, B., Ricard, S., Velázquez, J.A., Schmid, J., Minville, M., Caya, D., Chaumont, D., Ludwig, R., Turcotte, R., 2013. On the need for bias correction in regional climate scenarios to assess climate change impacts on river runoff. *Hydrol. Earth Syst. Sci.* 17 (3), 1189–1204. <https://doi.org/10.5194/hess-17-1189-2013>.
- Ng, J.L., Abd Aziz, S., Huang, Y.F., Mirzaei, M., Wayayok, A., Rowshon, M.K., 2019. Uncertainty analysis of rainfall depth duration frequency curves using the bootstrap resampling technique. *J. Earth Syst. Sci.* 128, 113. <https://doi.org/10.1007/s12040-019-1154-1>.
- NOAA NCEI (National Oceanic and Atmospheric Administration, National Centers for Environmental Information), 2014. *US 15 Minute Precipitation Data, Version 1.0*. 1970–2010.
- Ouali, D., Cannon, A.J., 2018. Estimation of rainfall intensity–duration–frequency curves at ungauged locations using quantile regression methods. *Stoch. Env. Res. Risk A.* 32, 2821–2836. <https://doi.org/10.1007/s00477-018-1564-7>.
- Perica, S., Martin, D., Pavlovic, S., Roy, I., St Laurent, M., Trypaluk, C., Unruh, D., Yekta, M., Bonnin, G., 2013. *Precipitation-Frequency Atlas of the United States (No.*

- Volume 9 Version 2.0: Southeastern States (Alabama, Arkansas, Florida, Georgia, Louisiana, Mississippi)). NOAA, National Weather Service, Silver Spring, MD.
- Prein, A.F., Rasmussen, R.M., Ikeda, K., Liu, C., Clark, M.P., Holland, G.J., 2017. The future intensification of hourly precipitation extremes. *Nat. Clim. Chang.* 7, 48–52. <https://doi.org/10.1038/nclimate3168>.
- Rampal, N., Gibson, P.B., Sherwood, S.C., Queen, L.E., Lewis, H., Abramowitz, G., 2025. Downscaling With AI Reveals the Large Role of Internal Variability in Fine-Scale Projections of Climate Extremes (arXiv:2507.06527) arXiv. doi:10.48550/arXiv.2507.06527.
- Sarhadi, A., Soulis, E.D., 2017. Time-varying extreme rainfall intensity-duration-frequency curves in a changing climate. *Geophys. Res. Lett.* 44, 2454–2463. <https://doi.org/10.1002/2016GL072201>.
- Scinocca, J.F., Kharin, V.V., Jiao, Y., Qian, M.W., Lazare, M., Solheim, L., Flato, G.M., Biner, S., Desgagne, M., Dugas, B., 2016. Coordinated global and regional climate modeling. *J. Clim.* 29, 17–35. <https://doi.org/10.1175/JCLI-D-15-0161.1>.
- Skamarock, W., Klemp, J., Dudhia, J., Gill, D., Barker, D., Wang, W., 2005. A Description of the Advanced Research WRF Version 2. UCAR/NCAR. <https://doi.org/10.5065/D6DZ069T>.
- Solaiman, T., 2011. Uncertainty Estimation of Extreme Precipitations Under Climate Change: A Non-Parametric Approach. *Electronic Thesis and Dissertation Repository*.
- Takhellambam, B.S., Srivastava, P., Lamba, J., McGehee, R.P., Kumar, H., Tian, D., 2022. Temporal disaggregation of hourly precipitation under changing climate over the Southeast United States. *Sci. Data* 9, 211. <https://doi.org/10.1038/s41597-022-01304-7>.
- Takhellambam, B.S., Srivastava, P., Lamba, J., Zhao, W., Kumar, H., Tian, D., Molinari, R., 2024. Artificial neural network-empowered projected future rainfall intensity-duration-frequency curves under changing climate. *Atmos. Res.* 297, 107122. <https://doi.org/10.1016/j.atmosres.2023.107122>.
- Tebaldi, C., Dorheim, K., Wehner, M., Leung, R., 2021. Extreme metrics from large ensembles: Investigating the effects of ensemble size on their estimates. *Earth Syst. Dynam.* 12 (4), 1427–1501. <https://doi.org/10.5194/esd-12-1427-2021>.
- Tung, Y., Wong, C., 2014. Assessment of design rainfall uncertainty for hydrologic engineering applications in Hong Kong. *Stoch. Env. Res. Risk A.* 28, 583–592. <https://doi.org/10.1007/s00477-013-0774-2>.
- Vidal, I., 2014. A Bayesian analysis of the Gumbel distribution: an application to extreme rainfall data. *Stoch. Env. Res. Risk A.* 28, 571–582. <https://doi.org/10.1007/s00477-013-0773-3>.
- Vu, T.M., Mishra, A.K., 2019. Nonstationary frequency analysis of the recent extreme precipitation events in the United States. *J. Hydrol.* 575, 999–1010. <https://doi.org/10.1016/j.jhydrol.2019.05.090>.
- Wang, X., Zhang, J., Shahid, S., Guan, E., Wu, Y., Gao, J., He, R., 2016. Adaptation to climate change impacts on water demand. *Mitig. Adapt. Strateg. Glob. Chang.* 21, 81–99. <https://doi.org/10.1007/s11027-014-9571-6>.
- Zhao, W., Abhishek, Kinouchi, T., 2022. Uncertainty quantification in intensity-duration-frequency curves under climate change: implications for flood-prone tropical cities. *Atmos. Res.* 270, 106070. <https://doi.org/10.1016/j.atmosres.2022.106070>.
- Zhao, W., Abhishek, Takhellambam, B.S., Zhang, J., Zhao, Y., Kinouchi, T., 2023. Spatiotemporal variability of current and future sub-daily rainfall in Japan using state-of-the-art high-quality data sets. *Water Resour. Res.* 59, e2022WR034305. <https://doi.org/10.1029/2022WR034305>.
- Zhao, W., Zhang, J., Abhishek, Ma, S., Liu, X., Xu, Y., Li, L., Fang, H., Liu, K., 2026. Event-level linkages between atmospheric circulation and anomalous precipitation types in a typical East Asian monsoon basin. *Geophys. Res. Lett.* 53 (2), e2025GL120098. <https://doi.org/10.1029/2025GL120098>.
- Zhu, J., 2013. Impact of Climate Change on Extreme Rainfall across the United States. *J. Hydrol. Eng.* 18, 1301–1309. [https://doi.org/10.1061/\(ASCE\)HE.1943-5584.0000725](https://doi.org/10.1061/(ASCE)HE.1943-5584.0000725).
- Zhu, J., Stone, M.C., Forsee, W., 2012. Analysis of potential impacts of climate change on intensity–duration–frequency (IDF) relationships for six regions in the United States. *J. Water Clim. Change* 3, 185–196. <https://doi.org/10.2166/wcc.2012.045>.